

2.1.4 The Size of a Set

Definition 4: Let S be a set. If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is a **finite set** and that n is the **cardinality** of S . The cardinality of S is denoted by $|S|$.

Example 12: Because the null set has no elements, it follows that $|\emptyset| = 0$.

Definition 5: A set is said to be infinite if it is not finite.

Example 13: The set of positive integers is infinite.

2.1.5 Power Sets

Many problems involve testing all combinations of elements of a set to see if they satisfy some property.

To consider all such combinations of elements of a set S , we build a new set that has as its members all the subsets of S .

Definition 6: Given a set S , the **power set** of S is the set of all subsets of the set S . The power set of S is denoted by $P(S)$.

Example 14: What is the power set of the set $\{0, 1, 2\}$?

Solution: $P(\{0, 1, 2\}) = \{\emptyset, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\}$ $\square$

Example 15: What is the power set of the empty set? What is the power set of the set $\{\emptyset\}$?

Solution: The empty set has exactly one subset, namely, itself. $\therefore P(\emptyset) = \{\emptyset\}$
The set $\{\emptyset\}$ has exactly two subsets, namely, $\emptyset$ and the set $\{\emptyset\}$ itself.
 $\therefore P(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$. $\square$

If a set has n elements, then its power set has 2^n elements $\rightarrow$ proof will be done later

~~Cartesian Product~~